A Technical Introduction to Stochastic gradient descent

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta L(\theta_t)$$

Section 1

Approximations in continuous time For small learning rate η {\textstyle \eta } stochastic gradient descent (w n) n ∈ N 0 {\textstyle (w_{n})_{n\in \mathbb {N} _{0}}} can be viewed as a discretization of the gradient flow ODE

Section 2

Linear regression Suppose we want to fit a straight line y ^ = w 1 + w 2 x {\displaystyle {\hat {y}}=w_{1}+w_{2}x} to a training set with observations ((x 1 , y 1) , (x 2 , y 2) … , (x n , y n)) {\displaystyle ((x_{1},y_{1}),(x_{2},y_{2})\ldots ,(x_{n},y_{n}))} and corresponding estimated responses (y ^ 1 , y ^ 2 , … , y ^ n) {\displaystyle ({\hat {y}}_{1},{\hat {y}}_{2},\ldots ,{\hat {y}}_{n})} using least squares. The objective function to be minimized is

Section 3

Second-order methods A stochastic analogue of the standard (deterministic) Newton–Raphson algorithm (a "second-order" method) provides an asymptotically optimal or near-optimal form of iterative optimization in the setting of stochastic approximation. A method that uses direct measurements of the Hessian matrices of the summands in the empirical risk function was developed by Byrd, Hansen, Nocedal, and Singer. However, directly determining the required Hessian matrices for optimization may not be possible in practice. Practical and theoretically sound methods for second-order versions of SGD that do not require direct Hessian information are given by Spall and others. (A less efficient method based on finite differences, instead of simultaneous perturbations, is given by Ruppert.) Another approach to the approximation Hessian matrix is replacing it with the Fisher information matrix, which transforms usual gradient to natural. These methods not requiring direct Hessian information are based on either values of the summands in the above empirical risk function or values of the gradients of the summands (i.e., the SGD inputs). In particular, second-order optimality is asymptotically achievable without direct calculation of the Hessian matrices of the summands in the empirical risk function. When the objective is a nonlinear least-squares loss

Section 4

Each {G(i,i)} gives rise to a scaling factor for the learning rate that applies to a single parameter wi. Since the denominator in this factor, G i = ∑ τ = 1 t g τ 2 {\textstyle {\sqrt {G_{i}}}={\sqrt {\sum _{\tau =1}^{t}g_{\tau }^{2}}}} is the ■2 norm of previous derivatives, extreme parameter updates get dampened, while parameters that get few or small updates receive higher learning rates. While designed for convex problems, AdaGrad has been successfully applied to non-convex optimization.

Section 5

This procedure will remain numerically stable virtually for all η {\displaystyle \eta } as the learning rate is now normalized. Such comparison between classical and implicit stochastic gradient descent in the least squares problem is very similar to the comparison between least mean squares (LMS) and normalized least mean squares filter (NLMS). Even though a closed-form solution for ISGD is only possible in least squares, the procedure can be efficiently implemented in a wide range of models. Specifically, suppose that Q i (w) {\displaystyle Q_{i}(w)} depends on w {\displaystyle w} only through a linear combination with features x i {\displaystyle x_{i}} , so that we can write ∇ w Q i (w) = – q (x i ′ w) x i {\displaystyle \nabla _{w}Q_{i}(w)=-q(x_{i}'w)x_{i}} , where q ( ) ∈ R {\displaystyle q()\in \mathbb {R} } may depend on x i , y i {\displaystyle x_{i},y_{i}} as well but not on w {\displaystyle w} except through x i ′ w {\displaystyle x_{i}'w} . Least squares obeys this rule, and so does logistic regression, and most generalized linear models. For instance, in least squares, q (x i ′ w) = y i – x i ′ w {\displaystyle q(x_{i}'w)=y_{i}-x_{i}'w} , and in logistic regression q (x i ′ w) = y i – S (x i ′ w) {\displaystyle q(x_{i}'w)=y_{i}-S(x_{i}'w)} , where S (u) = e u / (1 + e u) {\displaystyle S(u)=e^{u}/(1+e^{u})} is the logistic function. In Poisson regression, q (x i ′ w) = y i – e x i ′ w {\displaystyle q(x_{i}'w)=y_{i}-e^{x_{i}'w}} , and so on. In such settings, ISGD is simply implemented as follows. Let f (ξ) = η q (x i ′ w old + ξ ■ x i ■ 2) {\displaystyle f(\xi)=\eta q(x_{i}'w^{\text{old}}+\xi \|x_{i}\|^{2})} , where ξ {\displaystyle \xi } is scalar. Then, ISGD is equivalent to:

Section 6

Sign-based stochastic gradient descent Even though sign-based optimization goes back to the aforementioned Rprop, in 2018 researchers tried to simplify Adam by removing the magnitude of the stochastic gradient from being taken into account and only considering its sign. This results in a significantly lower communication cost of transferring gradients from workers to the parameter server. In this sense, it serves to better compress the gradient information, while having comparable convergence to standard SGD.

Section 7

where the parameter w {\displaystyle w} that minimizes Q (w) {\displaystyle Q(w)} is to be estimated. Each summand function Q i {\displaystyle Q_{i}} is typically associated with the i {\displaystyle i} -th observation in the data set (used for training). In classical statistics, sum-minimization problems arise in least squares and in maximum-likelihood estimation (for independent observations). The general class of estimators that arise as minimizers of sums are called M-estimators. However, in statistics, it has been long recognized that requiring even local minimization is too restrictive for some problems of maximum-likelihood estimation. Therefore, contemporary statistical theorists often consider stationary points of the likelihood function (or zeros of its derivative, the score function, and other estimating equations). The sum-minimization problem also arises for empirical risk minimization. There, Q i (w) {\displaystyle Q_{i}(w)} is the value of the loss function at i {\displaystyle i} -th example, and Q (w) {\displaystyle Q(w)} is the empirical risk. When used to minimize the above function, a standard (or "batch") gradient descent method would perform the following iterations:

Section 8

Further reading Bottou, Léon (2004), "Stochastic Learning", Advanced Lectures on Machine Learning, LNAI, vol. 3176, Springer, pp. 146–168, ISBN 978-3-540-23122-6 Buduma, Nikhil; Locascio, Nicholas (2017), "Beyond Gradient Descent", Fundamentals of Deep Learning : Designing Next-Generation Machine Intelligence Algorithms, O'Reilly, ISBN 9781491925584 LeCun, Yann A.; Bottou, Léon; Orr, Genevieve B.; Müller, Klaus-Robert (2012), "Efficient BackProp", Neural Networks: Tricks of the Trade, Springer, pp. 9–48, ISBN 978-3-642-35288-1 Spall, James C. (2003), Introduction to Stochastic Search and Optimization, Wiley, ISBN 978-0-471-33052-3